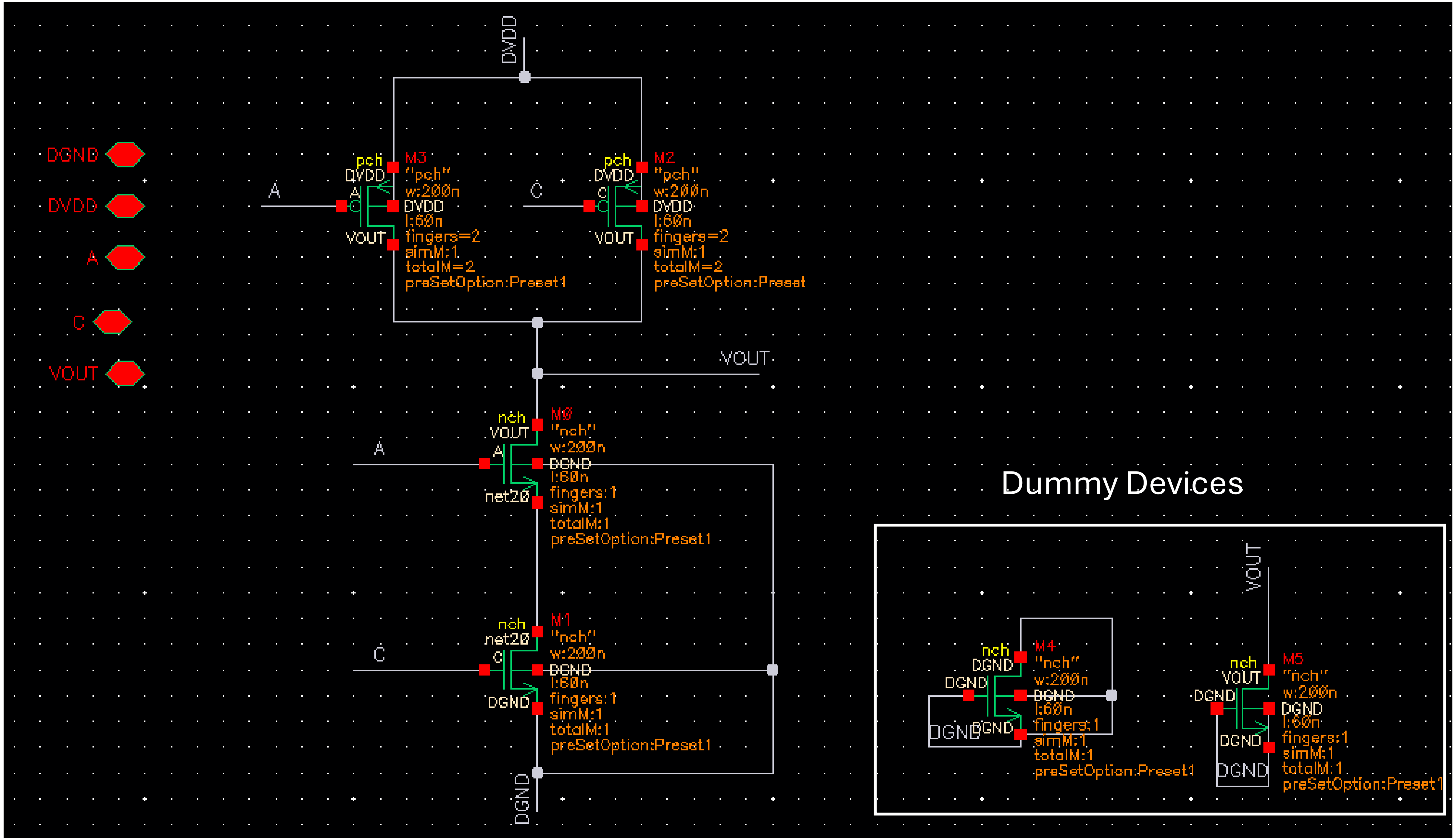
2-input NAND Gate Layout using TSMC 65nm Technology

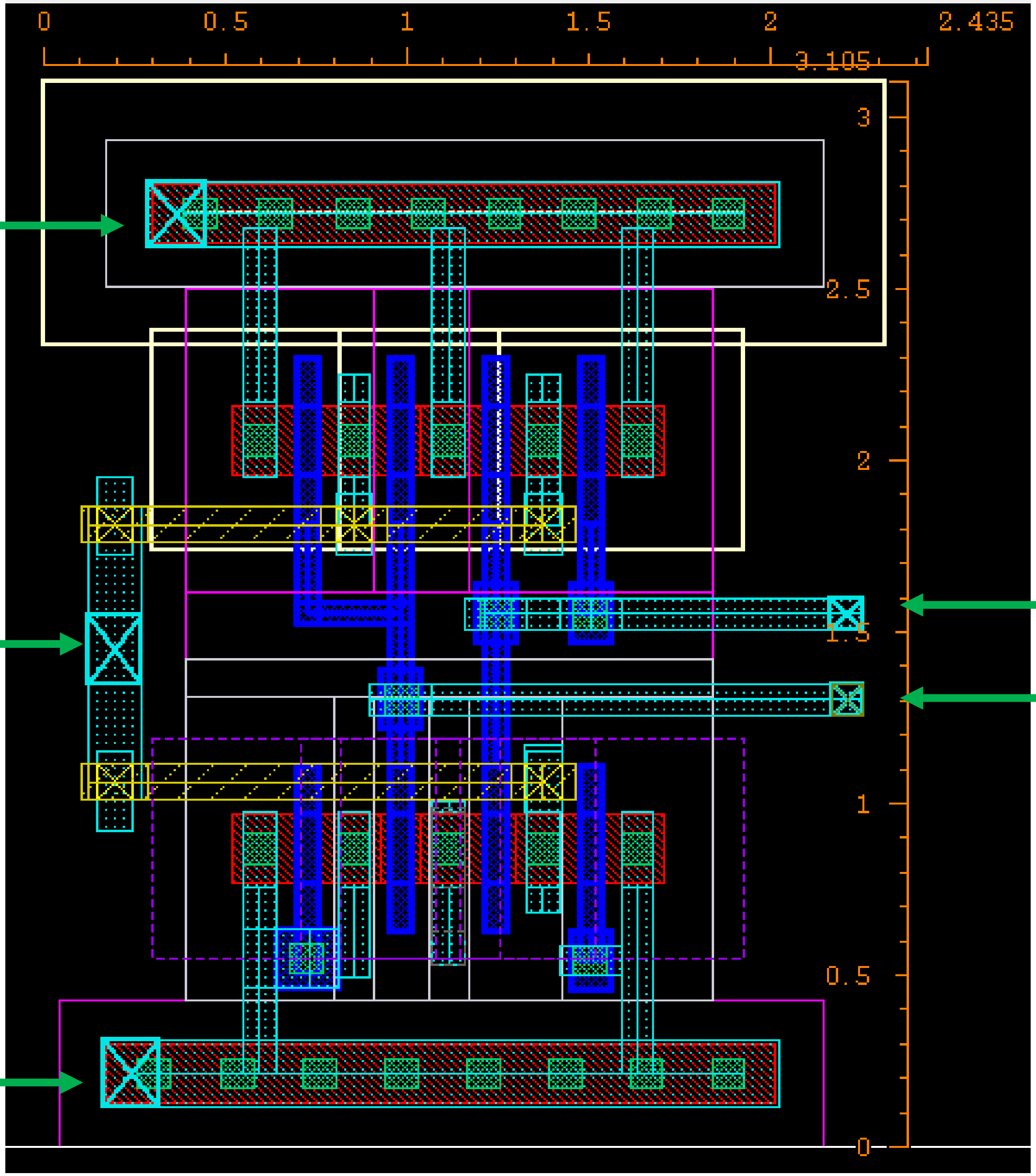
NAND Schematic



NAND Schematic Notes

- The Two PMOS devices have the same finger width as of NMOS .
- But since the current derive capability of NMOS is higher than that of PMOS because of mobility → the width of PMOS should be around the double that of the NMOS
- We use number of fingers to double the width of the PMOS devices
- We add two dummies with just one finger to make the number of NMOS devices = Number of PMOS devices which is 4 transistors.

NAND Layout



Supply Rail (DVDD)

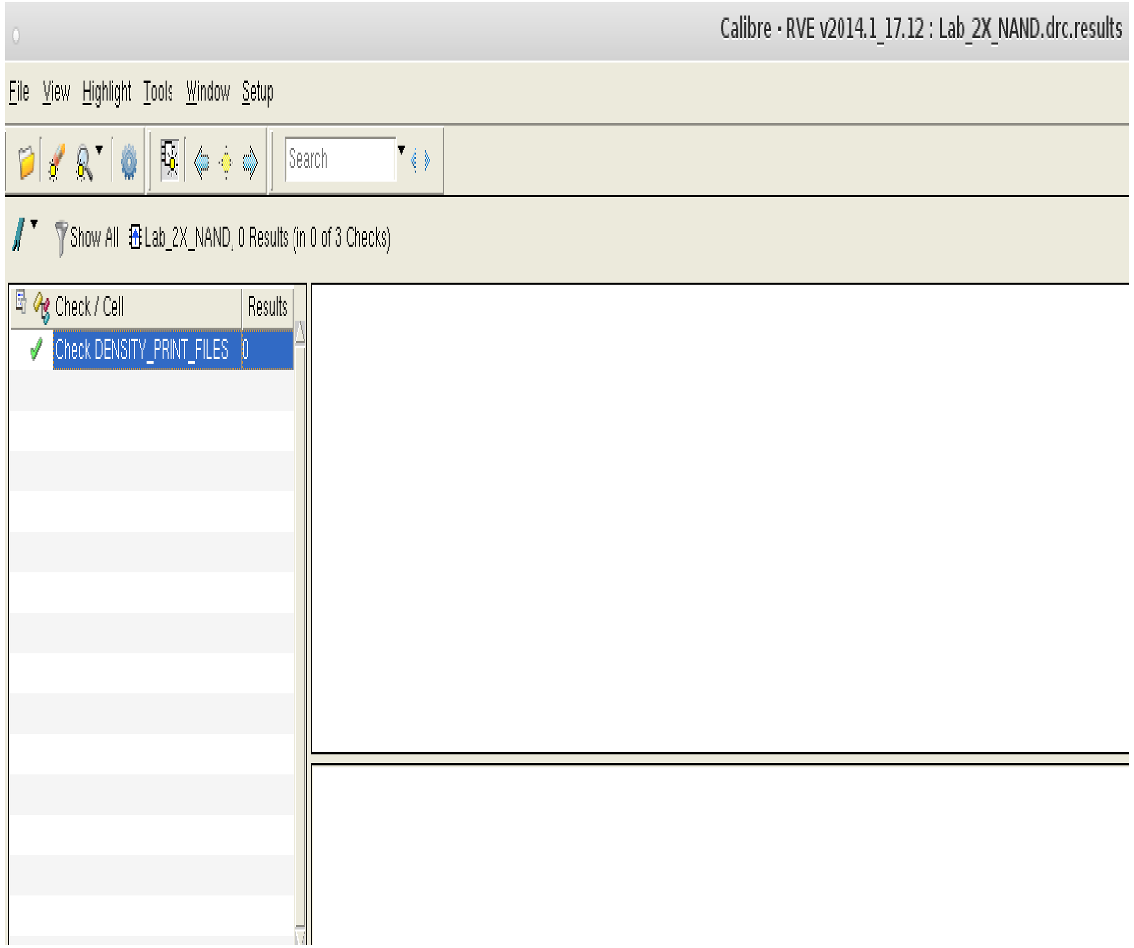
VOUT

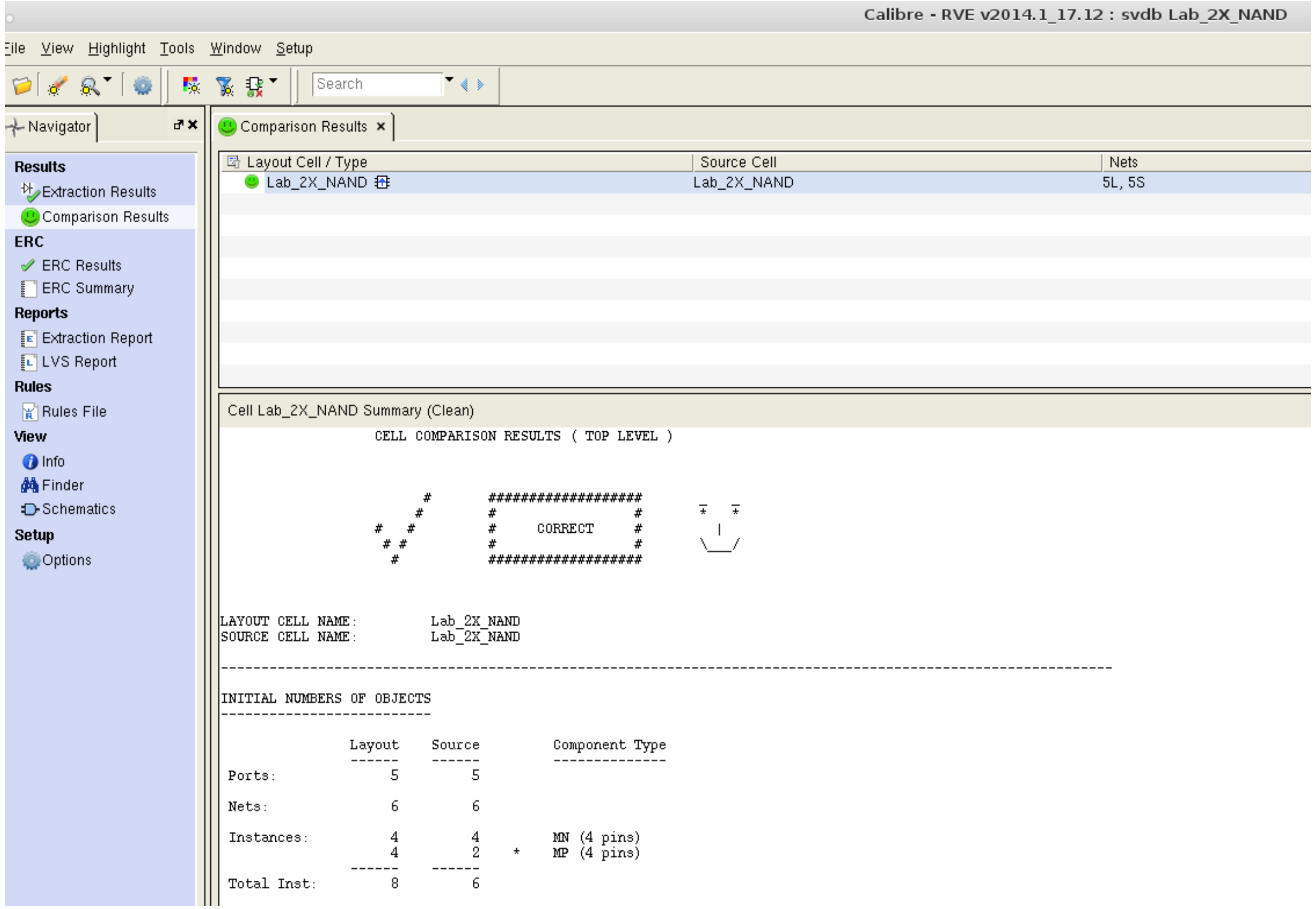
Ground Rail (DGND)

Input A

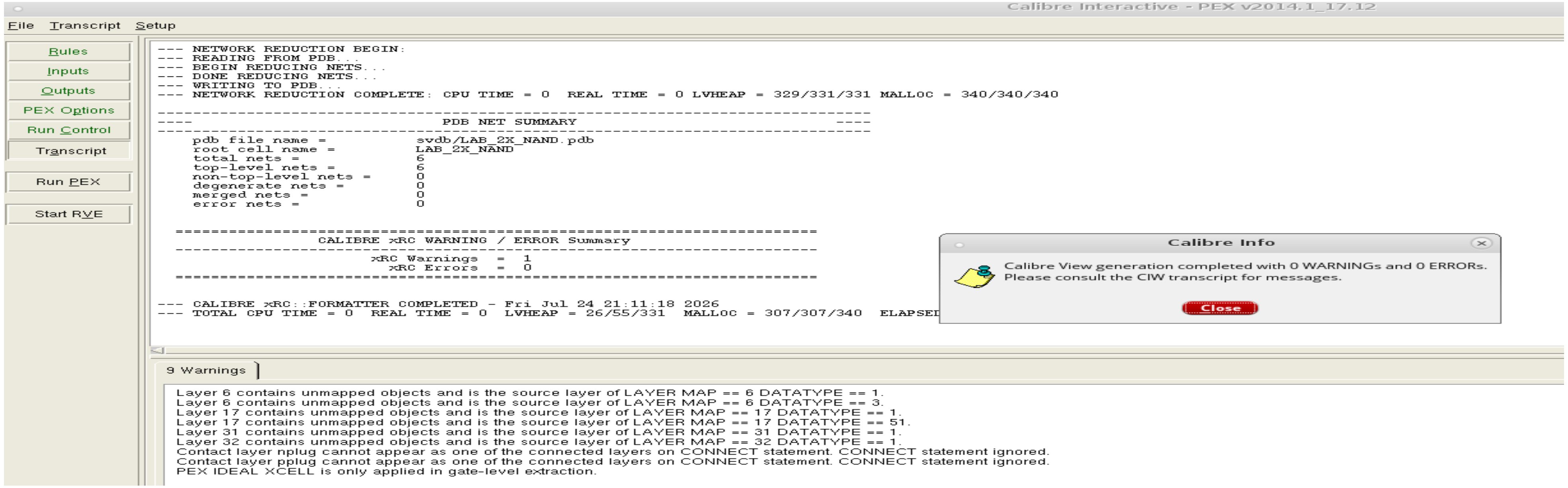
Input C

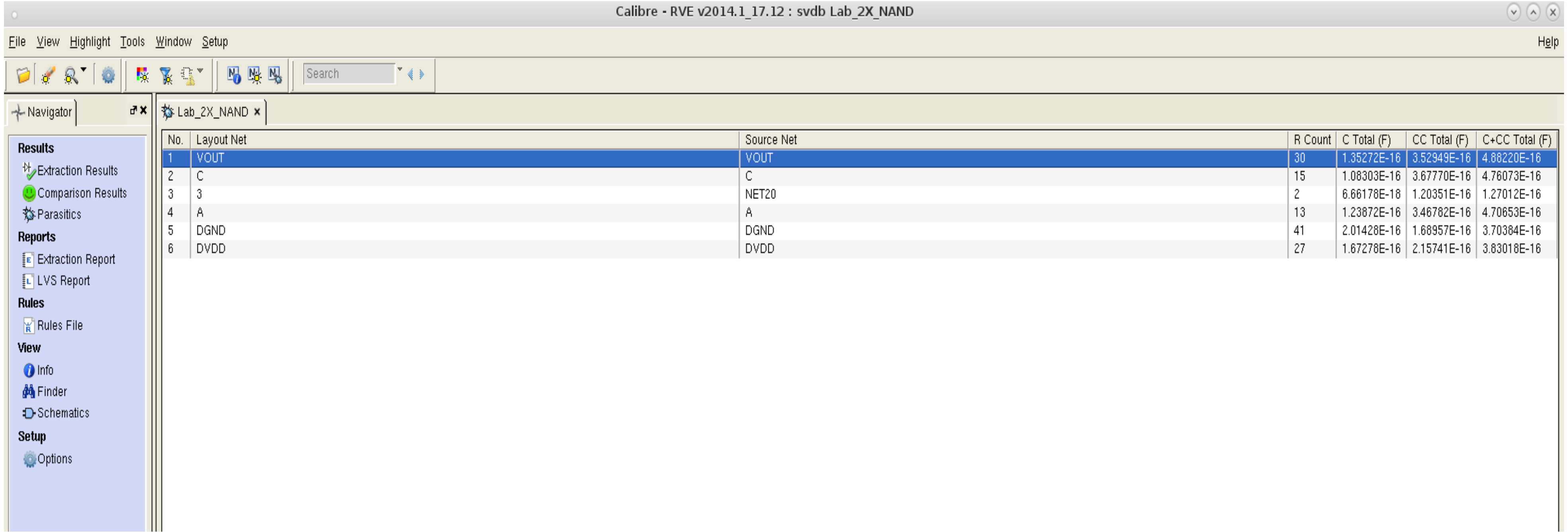
DRC and LVS


Calibre - RVE v2014.1_17.12 : Lab_2X_NAND.drc.results
File View Highlight Tools Window Setup
Search
Show All
Lab_2X_NAND, 0 Results (in 0 of 3 Checks)
Check / Cell
Results
Check DENSITY_PRINT_FILES
0


Calibre - RVE v2014.1_17.12 : svdb Lab_2X_NAND
File View Highlight Tools Window Setup
Search
Navigator
Comparison Results
Layout Cell / Type
Source Cell
Nets
Lab_2X_NAND
Lab_2X_NAND
5L, 5S
Results
Extraction Results
Comparison Results
ERC
ERC Results
ERC Summary
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Cell Lab_2X_NAND Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
LAYOUT CELL NAME: Lab_2X_NAND
SOURCE CELL NAME: Lab_2X_NAND
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
Ports: 5 5
Nets: 6 6
Instances: 4 4 MN (4 pins)
4 2 * MP (4 pins)
Total Inst: 8 6

PEX and Result Viewing Environment (RVE)

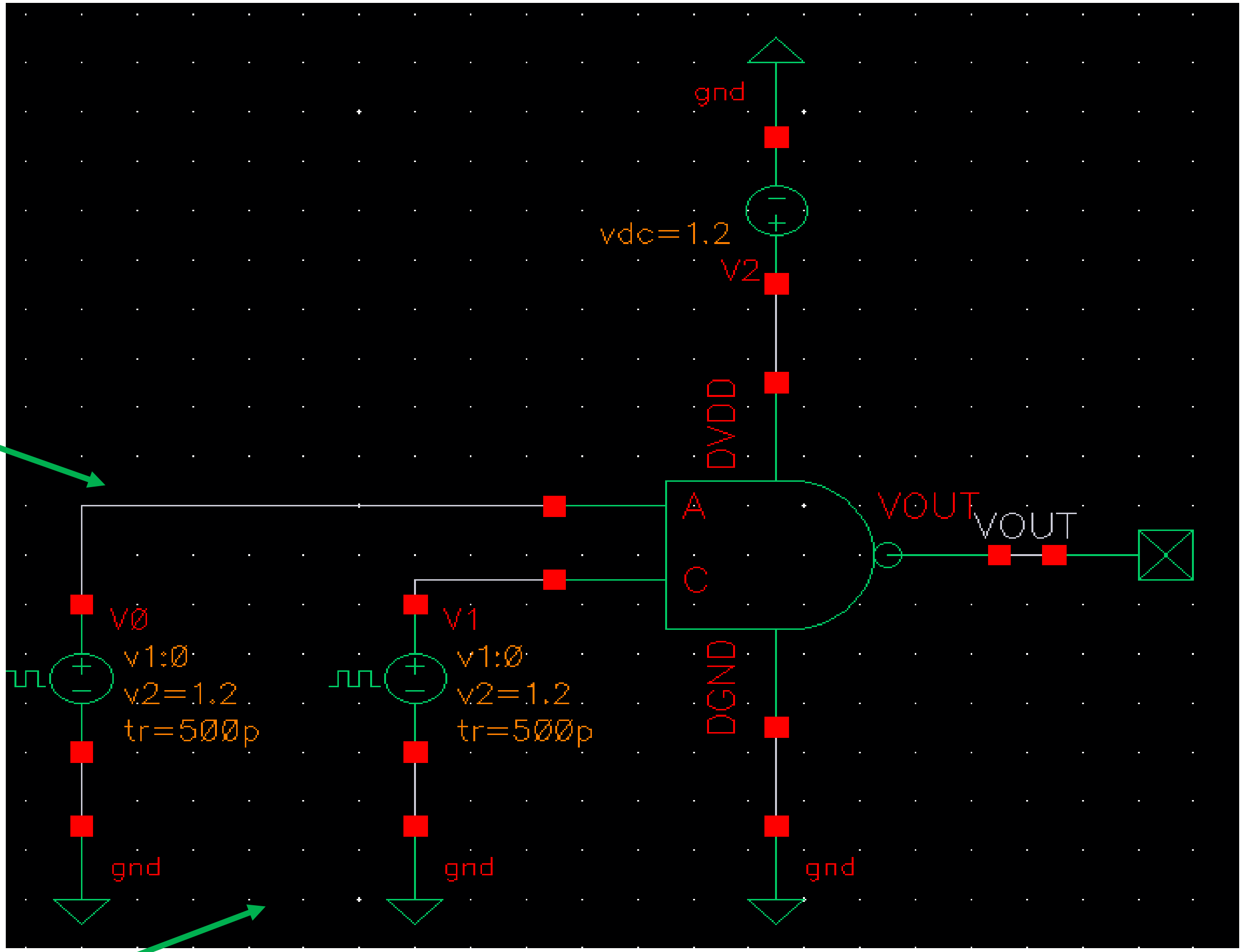




NAND Test Bench and Input Setup

DC voltage	1.2 V
AC magnitude	
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Voltage 1	0 V
Voltage 2	1.2 V
Period	20n s
Delay time	
Rise time	500p s
Fall time	500p s
Pulse width	10n s

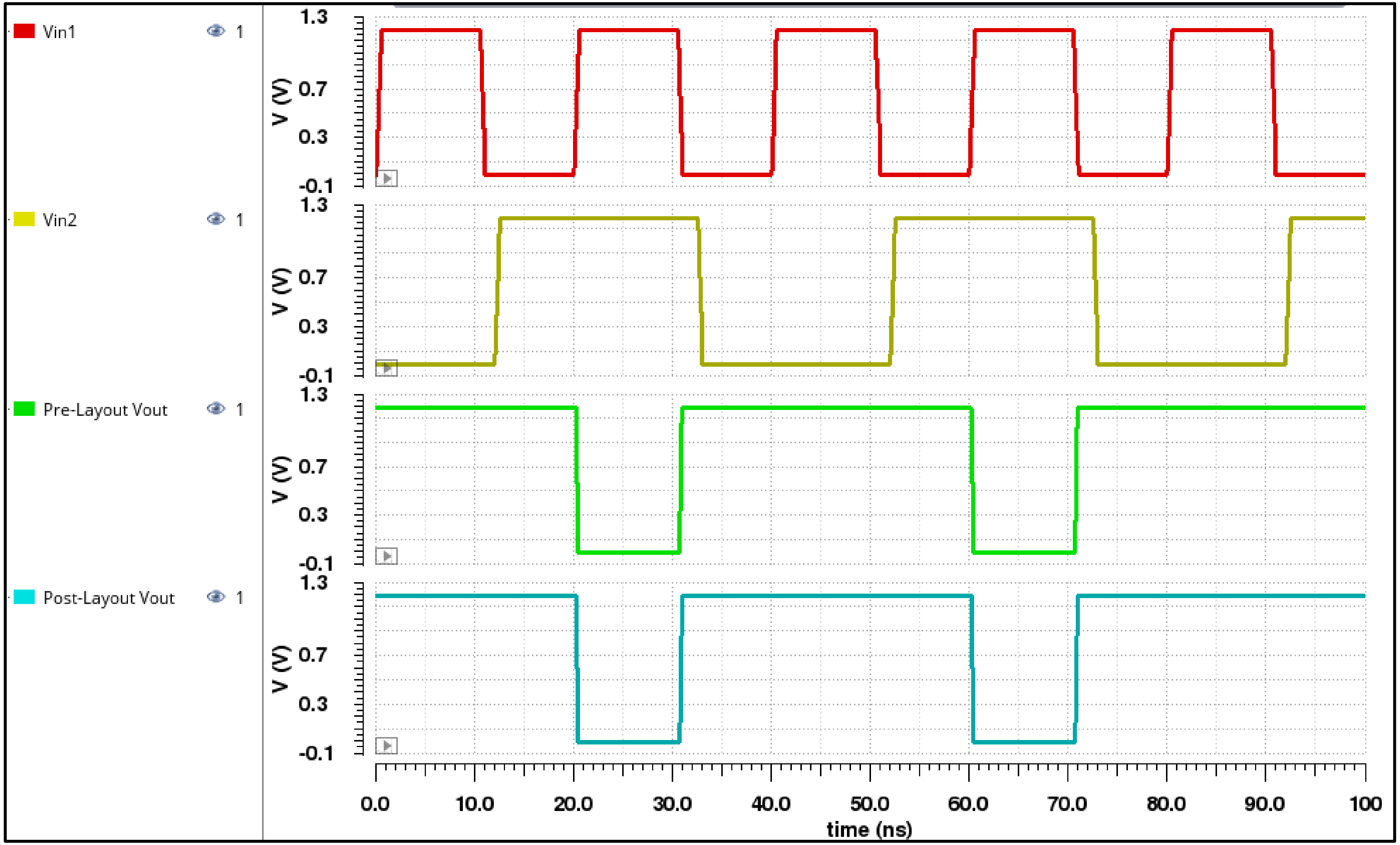
DC voltage	1.2 V
AC magnitude	
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Voltage 1	0 V
Voltage 2	1.2 V
Period	40n s
Delay time	12n s
Rise time	500p s
Fall time	500p s
Pulse width	20n s



Simulation Pre-Layout and Post-Layout results with Rise and Fall times

riseTime(VT("/VOUT") 0 nil 1....	74.2p
fallTime(VT("/VOUT") 1.2 nil 0...	79.94p

riseTime(VT("/VOUT") 0 nil 1....	86.38p
fallTime(VT("/VOUT") 1.2 nil 0...	93.85p

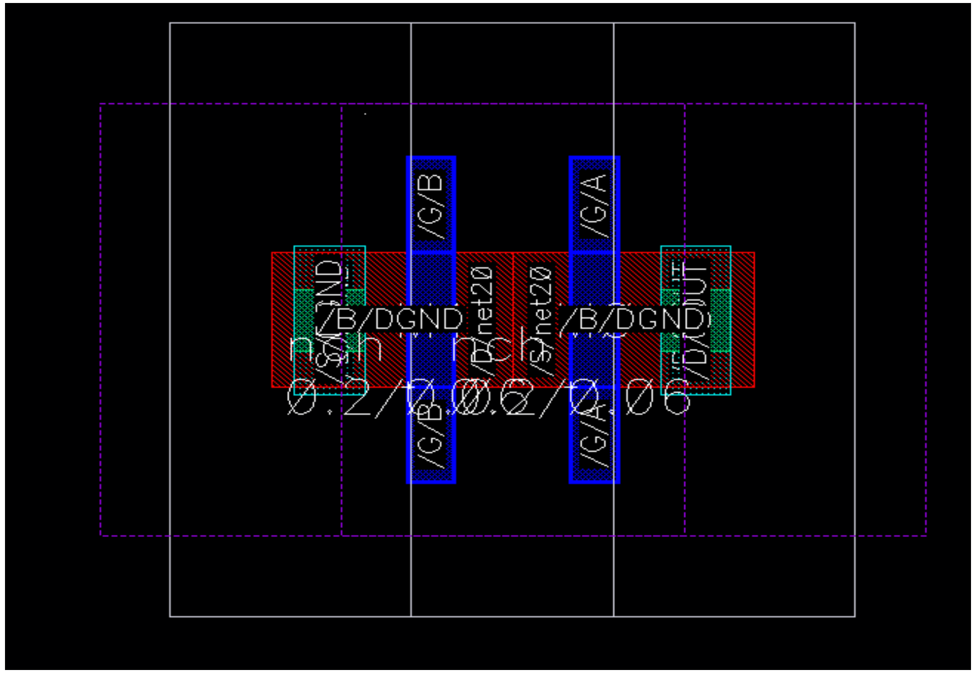


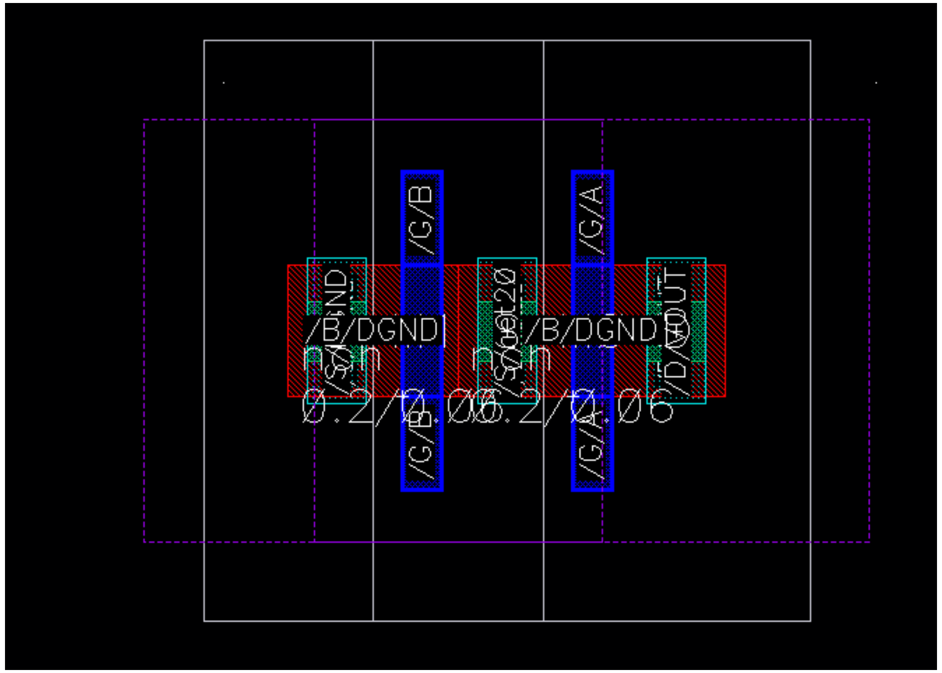
Layout Notes

NAND Layout Notes

- Latch-up (LUP) is avoided by the well
- When sharing for NMOS , the contact in the middle net disappears this will cause a problem of misalignment with the PMOS devices

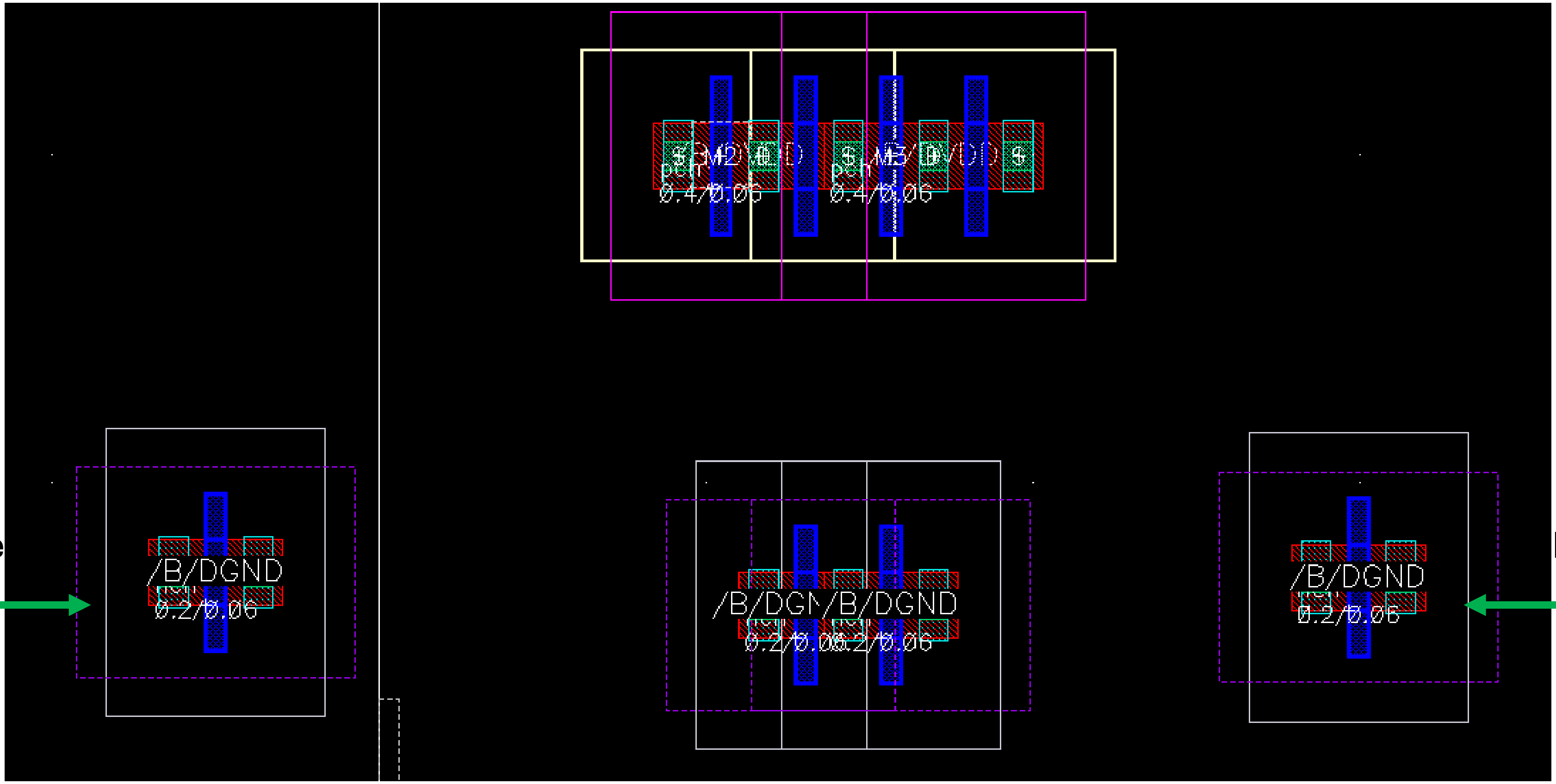
To get the contacts select the whole NMOS (the two devices) then right click and add the contact





NAND Layout Notes

- The PMOS devices are abutted to each other
- Now the NMOS devices with the dummy ones need to be abutted as well to make align connections between the two transistors

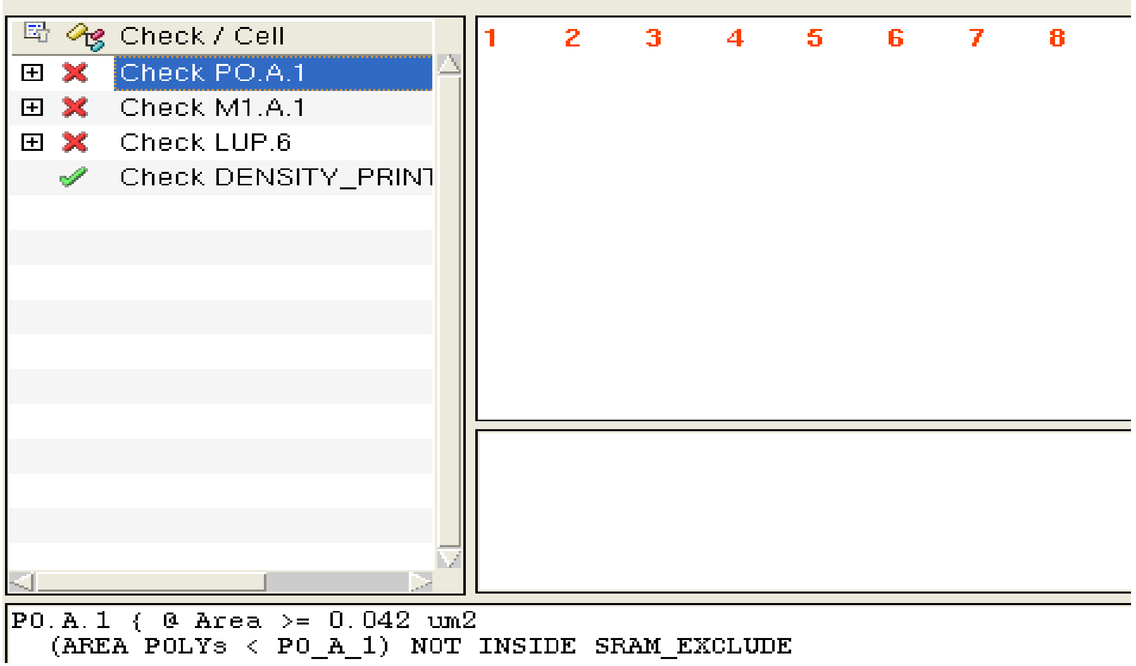


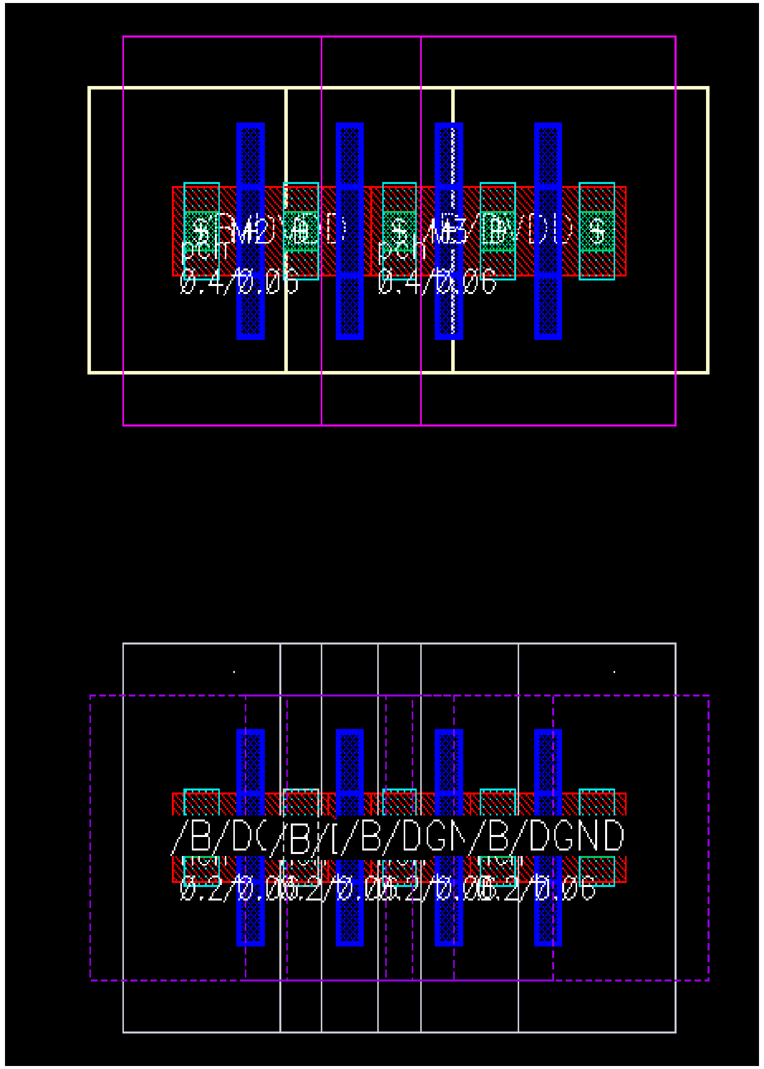
Dummy device

Dummy device

NAND Layout Notes

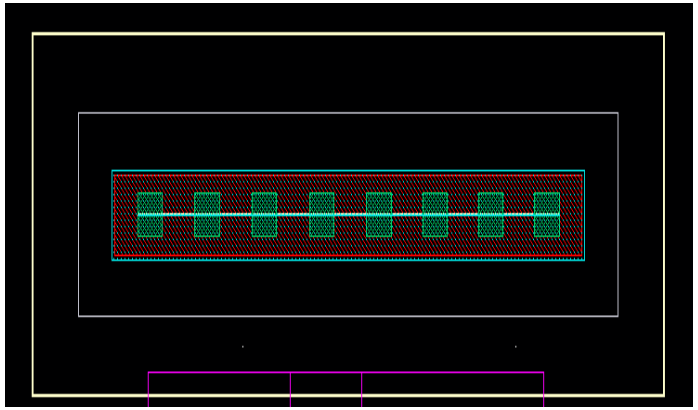
- After sharing the NMOS with dummies and PMOS , DRC is run
- There will be 3 errors
- One error for the LUP effect → solved by Nwell
- Two errors are for the area → PO Area and M1 Area should be > 0.042 um^2
- These area errors solved by increasing area or can be solved while routing

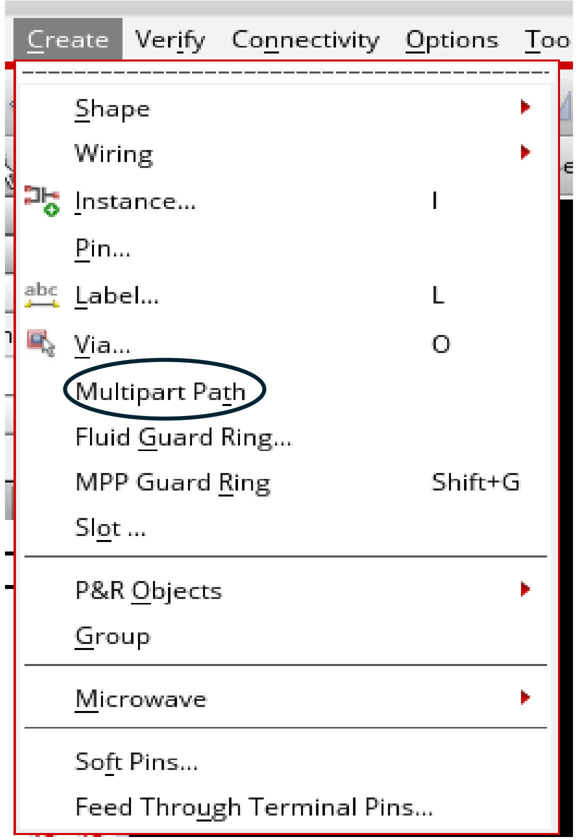


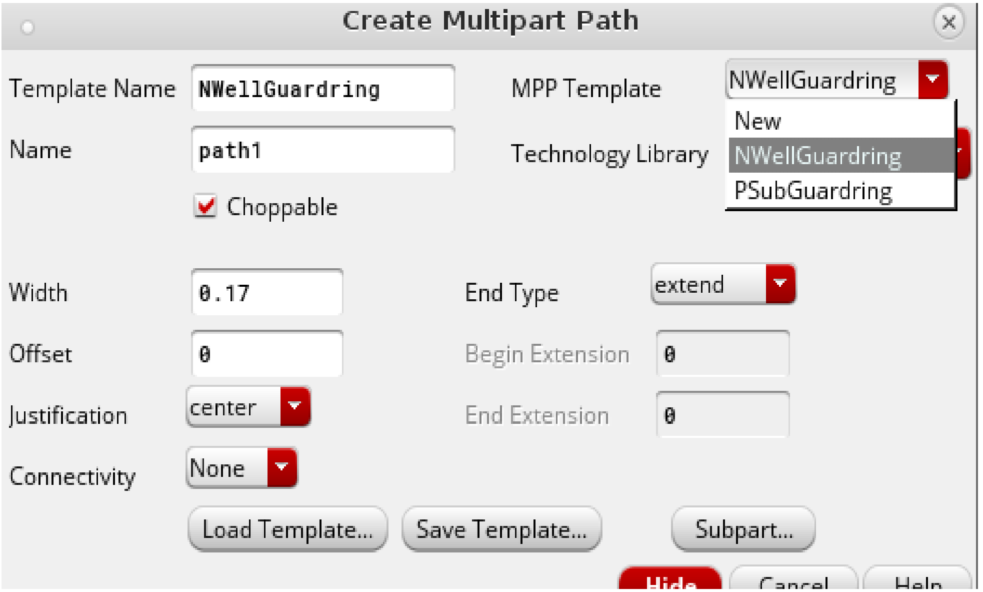


NAND Layout Notes

- To align to devices or components press A on the two sides to be aligned
- To create Vias → press p on the metal layer then press space → select the type of via you want with routing needed
- To create the VDD or GND rail → from create in the tool bar → multipart path → press f3 →from MPP template → for VDD ,select NWell guarding and for GND select PSub guarding

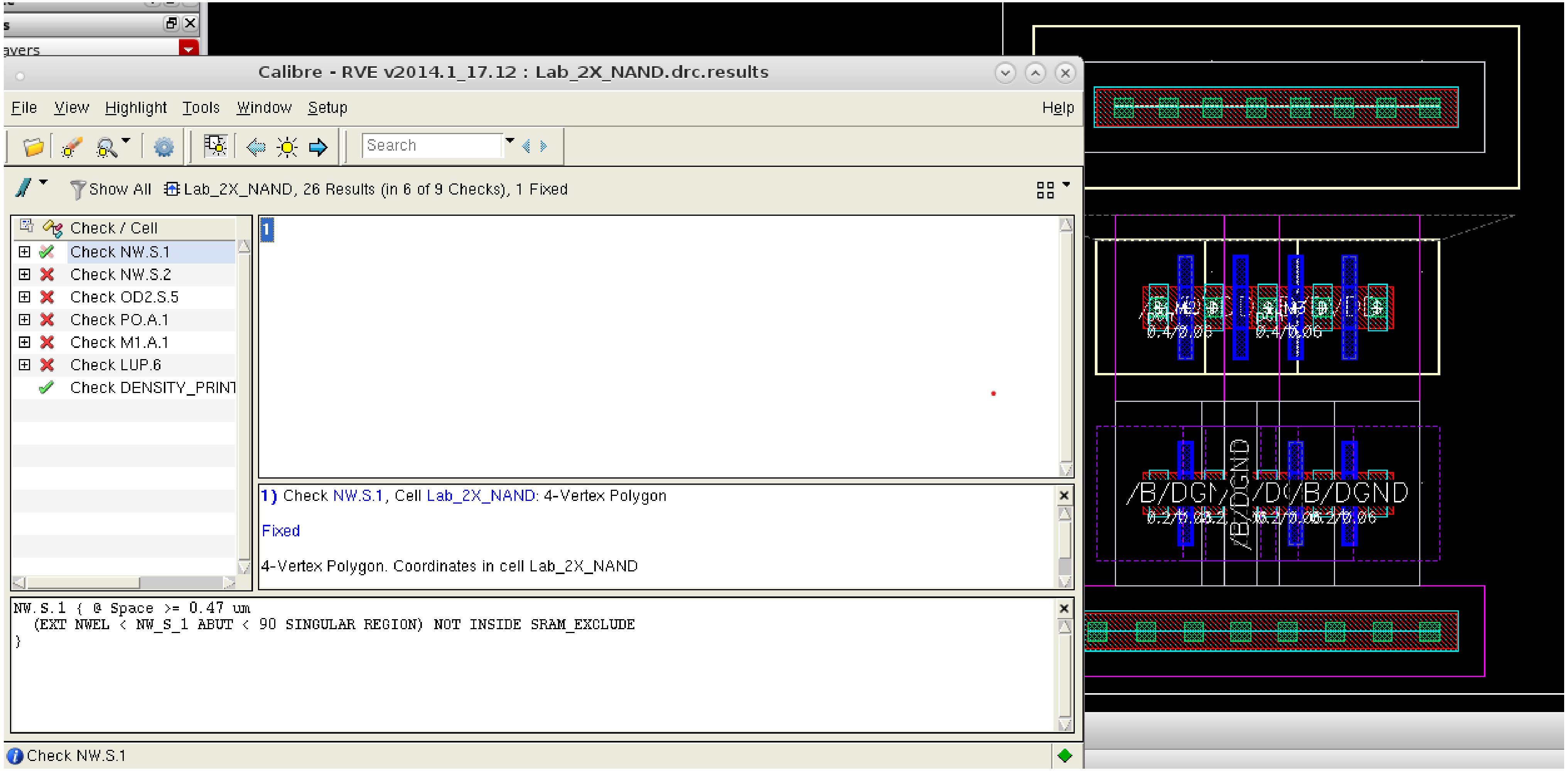






NAND Layout Notes

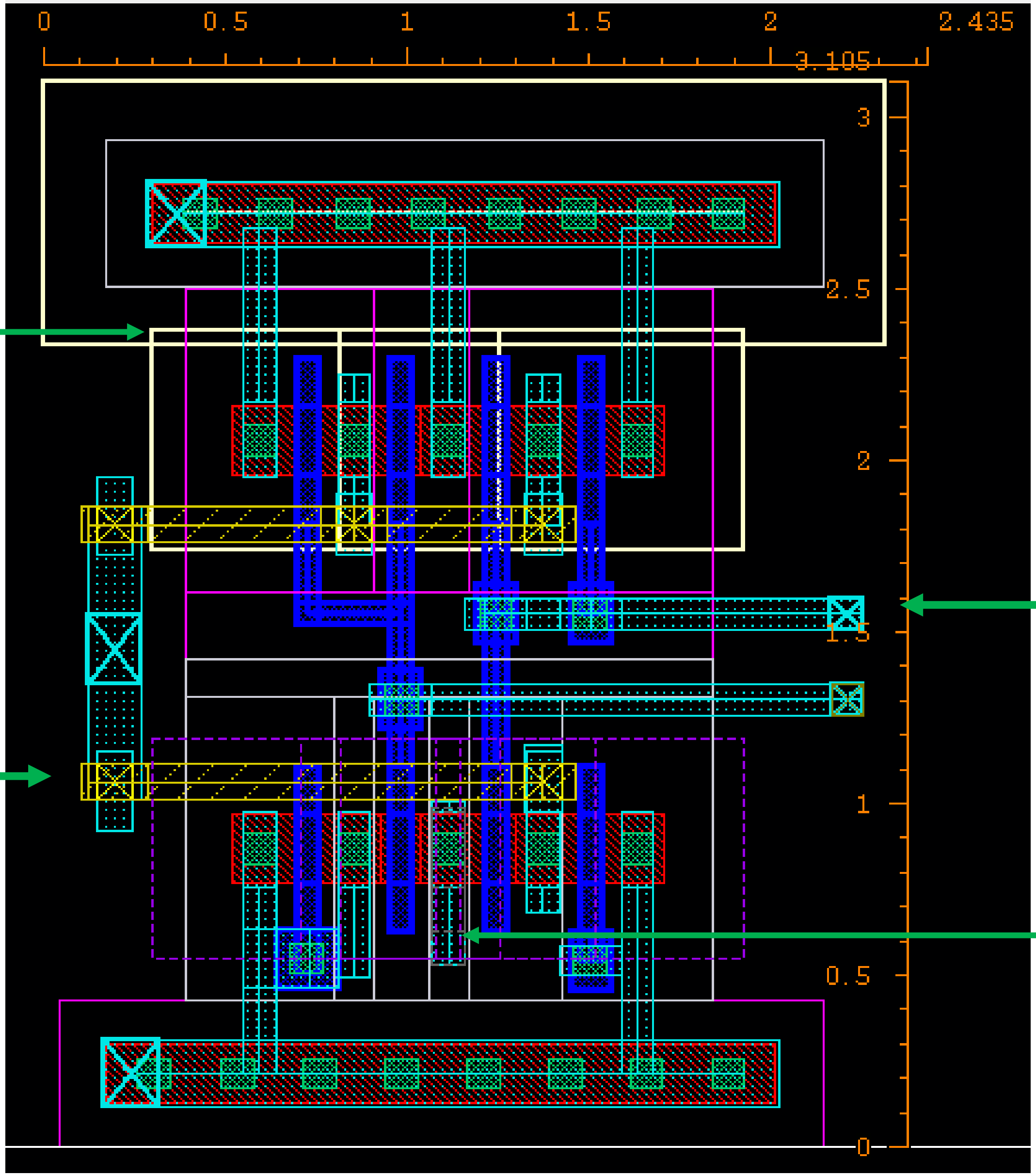
This DRC means the two Nwells should overlap on each other



NAND Layout Notes

The NWELL of DVDD must overlap with NWELL of PMOS

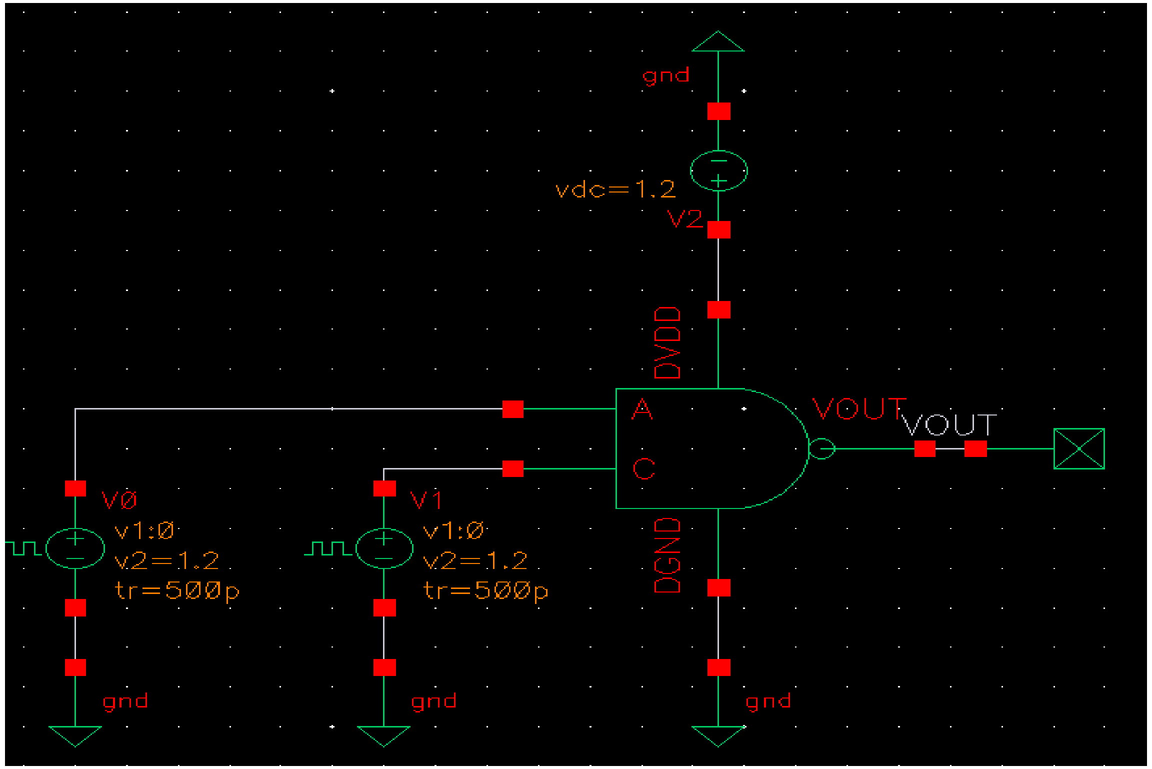
M2 was used to try to achieve Manhattan routing technique

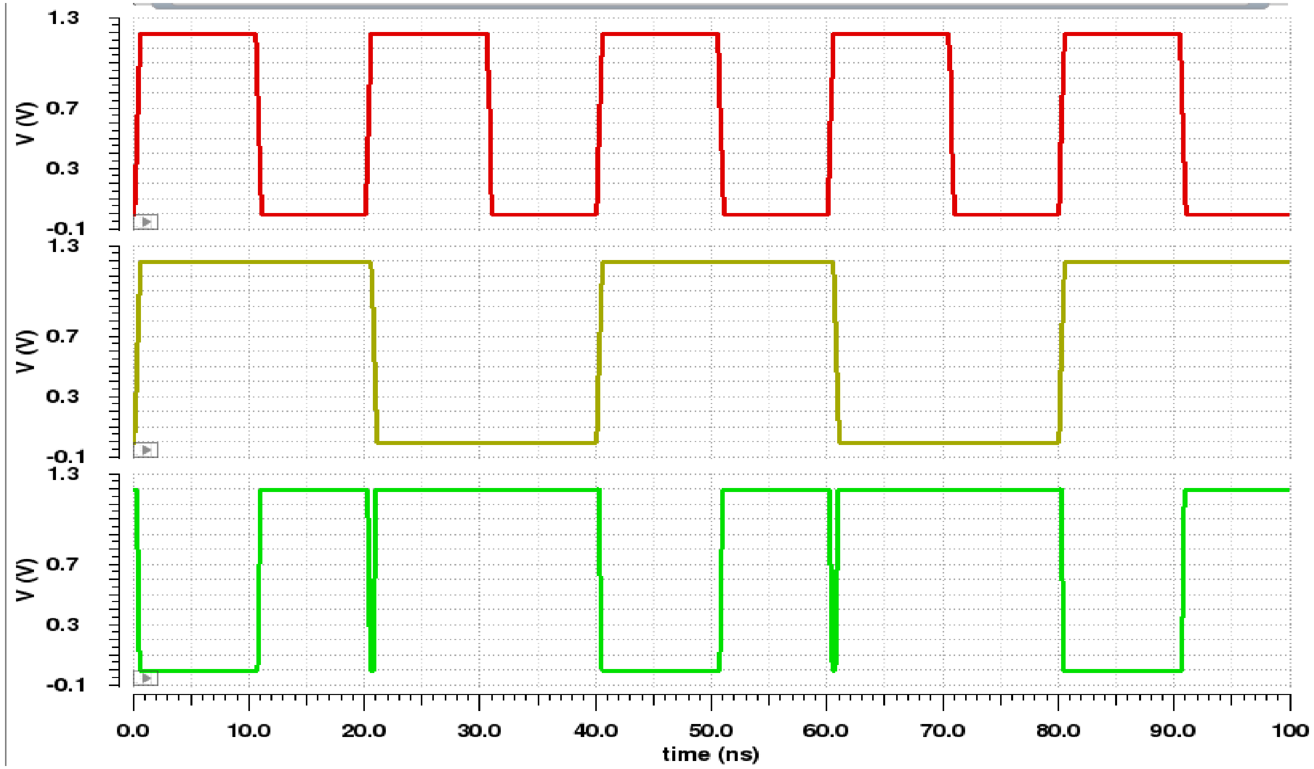


It would be better if the horizonatal routing was in M2 layer . The problem I made was that I didn't increase the distance between the devices PMOs and NMOS → Vias of M2 made problem of spacing (should be considered)

- This is the shared node between the two NMOS devices (net20)
- M1 of this node is increased to pass the DRC which is Area of M1 > 0.042 um^2

NAND Layout Notes





- To avoid race condition when input1 goes from low to high and input2 from high to low → put delay in input2 of 12 ps